Evaluating the Accuracy of Horner's Method

Jaden Krause

September 15, 2026

1 Introduction

Evaluating polynomials can be a simple task, depending on the size of the expression. The higher the degree of the function, the more tedious solving it can become. This is also an important consideration when using a computer to solve polynomials. The speed of solving the function can be improved if the step-by-step operations are reduced to simple expressions. This is where Horner's Method applies itself. This method successively factors out the independent variable until the expression is reduced to single-degree, "nested" expressions. This setup is advantageous when using a computer to evaluate polynomials, as it is easy to repetitively evaluate the simplified expressions using code. In this report, I give a comparison between evaluating polynomials normally and using Horner's Method, and the respective errors I obtained due to floating-point error and loss of significance.

2 Problem 1

$$P(x) = 1 + x + ... + x^{50}$$

$$x = 1.00001$$

I evaluated the above expression at the indicated value of 'x' using Python. I obtained the following values:

Method	Result
Polynomial	51.012752082745230
Horner's	51.012752082749991

Table 1: Problem 1 Results

Error Type	Result
Absolute	4.8e-12
Relative	9.3e-12%

Table 2: Error Approximations 1

These results show that Horner's Method yields a very close approximation of the exact polynomial evaluated at the point. The calculated results show no discrepancy until the 12th decimal place.

3 Problem 2

$$P(x) = 1 - x + x^2 - x^3 + ... + x^{98} - x^{99}$$

$$x = 1.00001$$

I evaluated the above expression at the indicated value of 'x' using Python, obtaining the following values:

Method	Result
Polynomial	-0.000500245079647
Horner's	-0.000500245079648

Table 3: Problem 2 Results

Error Type	Result
Absolute	1.7e-16
Relative	3.4e-11%

Table 4: Error Approximations 2

These results further support the validity of Horner's Method by showing very similar results, having no disagreement until the 15th decimal place.

4 A Discussion of Floating-Point Error and Loss of Significance

When using code to compute expressions like this, it is important to consider possible errors and how they affect the computation. Because computers have finite precision, rounding error can be compounded by certain calculations and terminate in imprecise results. (Sauer, Timothy, *Numerical Analysis*). These errors cause a loss of significant digits in the answer. In the example of the problems solved above, repeated multiplication combined with the computer's finite precision gave the errors found. Fortunately, we didn't experience any loss of significant digits, as our independent variable possessed 6, and our answers only diverged at much higher numbers of precision.

Occasionally, expressions can be manipulated to yield more precise answers. This is especially evident in trigonometric expressions and expressions involving square roots.

5 Different Solution Types

When solving differential equations, multiple types of solution can suit the purpose and provide different advantages and disadvantages. Explicit, implicit, graphical, and numerical solutions provide distinct benefits when it comes to the computation, precision, and ease of use for your data.

6 Conclusion

In this report, we discussed the usefulness of Horner's Method in evaluating high-degree polynomials when using a computer. We looked at two example problems and the absolute and percentage error between Horner's Method and exact polynomial evaluation. We discussed the inherent errors in computational evaluation of complicated expressions, and we briefly addressed the range of solutions when solving both differential equations and complex expressions.

7 Bibliography

Sauer, Timothy (2017). *Numerical Analysis*. 3rd ed. Pearson. ISBN: 9780134696454